Term Project – Option 1

# Highway toll removal in Catalonia and severe traffic accidents

Overview of background, assignment, dataset and preparation for the meeting with Marta Castellano · Data Science and AI for Industry (2400200), UPC · October 2026

**In short.** On 1 September 2021 tolls were removed on the AP-7, AP-2, C-32 North and C-33 in Catalonia. The task is to find out whether severe traffic accidents (death or serious injury) changed character after this, and to build a classifier that separates accidents before and after that date. The classifier must then be interpreted: *which* characteristics separate the two periods? The hard part is not getting high accuracy, but separating the effect of the toll removal from everything else that changed between 2010 and 2023.

## 1. Background: what happened to the toll roads?

Abertis operated the toll motorways in Catalonia under concessions for around 50 years. When the concessions expired, the Spanish State (AP-7, AP-2) and the Generalitat (C-32, C-33) chose not to renew them. Maintenance moved to public budgets and the roads became free. In total, roughly 550–600 km of motorway, including ramps, became toll-free on the same day.

### Timeline

| Date | Event | Why it matters for the analysis |
|---|---|---|
| 2 Sep 2018 | Mandatory diversion of heavy vehicles from the N-340/N-240 to the AP-7/AP-2 (with subsidised tolls) in the Tarragona region. | Heavy traffic shifted *before* 2021. An early change that is not the toll removal. |
| 1 Jan 2020 | AP-7 South (Tarragona–Alacant) becomes free, including the Vila-seca–Alcanar stretch in Catalonia. | A "toll effect" already starts in 2020 on the southern AP-7. Can be separated using the km point (pk). |
| Mar–Jun 2020 | Covid lockdown. 2020 has ~30% fewer accidents than 2019. | Extreme anomaly that distorts both the trend and the "before" class. |
| **1 Sep 2021** | **Toll-free:** AP-7 Vila-seca–La Jonquera (French border), AP-2 Zaragoza–El Vendrell (Penedès), C-32 North (Montgat–Blanes, Maresme), C-33 Barcelona–Montmeló. | The intervention itself. The before/after split in the assignment. |
| After 2021 | Tolls remain on the C-32 South (Castelldefels–El Vendrell, until 2039) and the C-16 (Terrassa–Manresa, Vallvidrera and Cadí tunnels). | Roads that did *not* become free – a possible comparison group. |

### What the media and the authorities reported

- Traffic on the AP-7 rose 20.5% in September 2021 compared with September 2019; the C-33 rose 46%. The authorities pointed out that much of the increase was traffic moving over from parallel roads, not necessarily new traffic.
- Sept. 2019 vs. Sept. 2021: accidents involving a truck on the four roads rose from 11 to 18. Fatal accidents on the parallel N-II and N-340 roughly halved (Jan–Sep 2019 vs. 2021).
- Servei Català de Trànsit (2024): injury accidents on the AP-7 are 28% higher than before the toll removal. Traffic at Martorell rose from about 80,000 to 106,000 vehicles/day. Accidents and traffic on alternative roads fell.
- The hypothesis you can test: traffic (especially heavy and long-distance traffic) moved from two-lane roads such as the N-II/N-340 onto the motorways. This should give *more* severe accidents on the former toll roads, *fewer* on the parallel roads, and a different accident profile (e.g. fewer head-on collisions, more rear-end and congestion-related crashes).

## 2. The assignment

| | |
|---|---|
| Question | Did the removal of highway tolls on 1 September 2021 change the characteristics of severe traffic accidents? |
| Part 1 – EDA | Explore whether accident characteristics changed before vs. after. |
| Part 2 – Classifier | Predict whether an accident happened before or after 1 Sep 2021 (the assignment text says "2025" – a typo). |
| Part 3 – Interpretation | Explain which variables drive the classification. This is the part that actually answers the question. |
| Critical points (Marta) | Class imbalance · changes over time · derived features · validation strategy · appropriate metrics. |
| Report | 11–14 pages, 11 pt, single spacing: Executive summary (1 p), Project structure (1 p), Data & EDA (3–4 p), Model (3–4 p), Results (2–3 p), Next steps (1 p). R code in an appendix. |
| Deadlines | Review meeting with the professor in October (Google Meet). Submission Friday 27 November, 23:59, in Atenea. The project counts for 25% of the grade. |

## 3. The dataset

| | |
|---|---|
| Source | "Accidents de trànsit amb morts o ferits greus a Catalunya", Dades Obertes Catalunya (Departament d'Interior i Seguretat Pública). Dataset ID rmgc-ncpb. |
| Updates | Annual. Last updated December 2024. The file covers **2010–2023**, not 2024 as the assignment states. |
| Size | 24,478 accidents × 58 variables. One row = one accident with at least one death or serious injury, both urban and interurban. |
| Class balance | Before: 20,643 (84.3%). After: 3,835 (15.7%). The after-period is only 28 months. |
| Files in the Project folder | Original CSV · *accidents_EN.csv* (translated) · *Kodebok_variabler.md* · *01_first_look.R* |

### Variable groups

| Group | Examples (English names) | Comment |
|---|---|---|
| Time | year, date, hour, time_of_day, weekday_group | Encode time directly – must be excluded from the classifier (leakage). |
| Location | road, km_point, municipality, county, province, zone | *road* is the key to identifying the former toll roads. |
| Parties involved | n_units, n_motorcycles, n_heavy_vehicles, n_pedestrians, n_bicycles, n_other_units … | Count variables. Fine as features. |
| Outcome | severity, n_dead, n_serious_injured, n_victims | Severity. Consider whether they are inputs or outcomes. |
| Accident type | accident_type, accident_subtype, hit_and_run | The core of "typology". |
| Road | road_type, speed_limit, intersection, junction_type, directions, road_owner … | See data quality below – *road_type* is unreliable. |
| Conditions | weather, surface, light, fog, wind, infl_* | infl_* = police assessment of whether the factor influenced the accident. |

# 4. First findings from the data

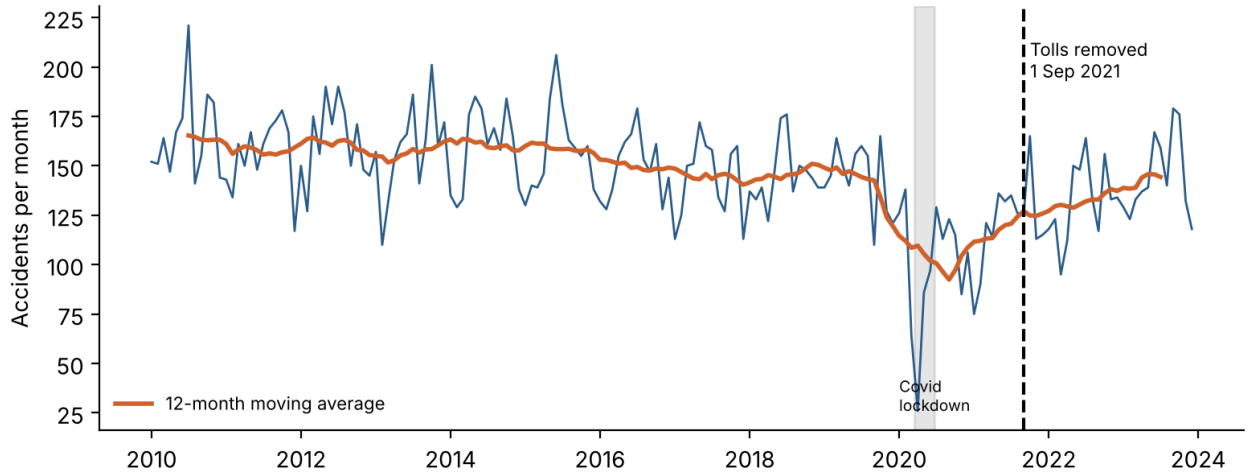



Figure 1. Severe accidents per month. Clear downward trend 2010–2019, Covid gap in 2020 and seasonal variation. The after-period sits at a level approaching pre-Covid.

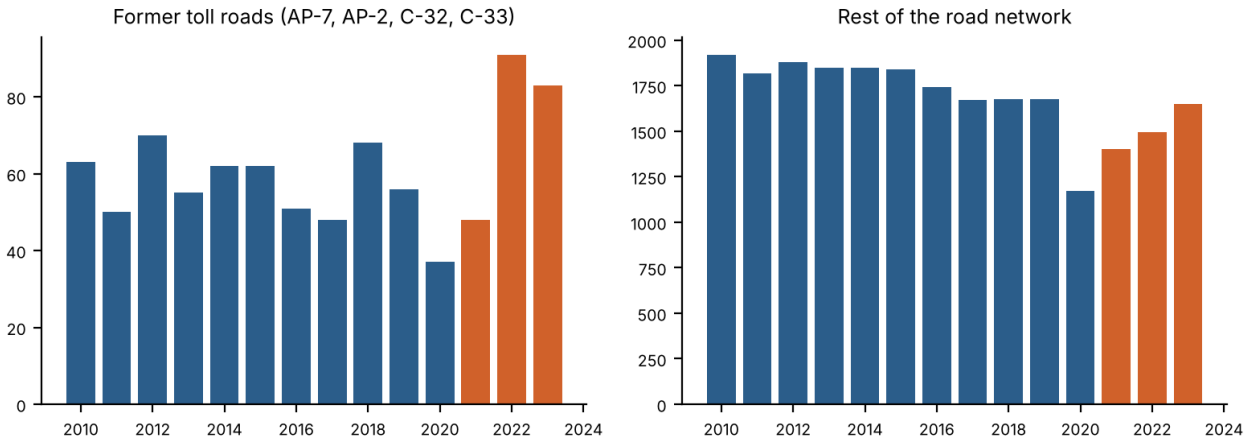



Figure 2. Accidents per year on former toll roads (left) vs. the rest of the network (right). Orange = 2021 and later. The former toll roads jump from about 50–70 to 91 and 83 in 2022–23, while the rest does not increase correspondingly. Note the small n (≈ 850 accidents in total on these roads).

## Change in shares before → after

| Characteristic | Before | After | Interpretation / caveats |
|---|---|---|---|
| Share of fatal accidents | 12.6% | 12.7% | Overall severity is unchanged. |
| Motorcycle involved | 38.1% | 41.3% | Could be a trend, not the tolls. |
| Bicycle involved | 8.1% | 10.1% | Trend (cycling is increasing). |
| "Other units" involved | 2.1% | 6.6% | Jumps from 2020 – likely e-scooters (VMP). Pure time trend. |
| accident_type = "Other" | 5.1% | 10.1% | Rises steadily from 2010 – probably the same cause/recording practice. |
| Pedestrian hit | 21.6% | 18.3% | Trend. |
| Urban zone | 54.5% | 57.2% | Small shift towards urban. |
| Accident on former toll road | 3.1% | 5.1% | The most direct signal of a toll effect. |

The point: the largest differences across the whole dataset are driven by time trends (e-scooters, bicycles, recording practice), not by the tolls. A classifier trained on all accidents will learn these, and the interpretation will be wrong unless you deal with it.

# 5. Data quality and pitfalls

- **road_type is not consistent over time.** "Autopista" is used almost only in 2013–2015; the AP-7 is coded as "Conventional road" in the other years. Use *road* (road name) to define motorways/former toll roads, not road_type.
- **Hidden missing values:** 999 in speed_limit (1,715 rows), 999999 in km_point (12,750 rows), "Unspecified" and NA in many categories.
- **Missingness increases over time:** e.g. gradient (51% → 66%) and directions (12% → 30%). The NA share alone can then separate before/after – another form of leakage.
- **Structurally missing:** road_owner, junction_type and priority_control are NA mostly for urban accidents. Do not impute blindly; consider a separate category or analysing urban and interurban separately.
- **Leakage from time variables:** year and date must be excluded. The same applies to variables that almost only appear in one period (n_other_units).
- **Outcome variables:** severity, n_dead and n_victims describe the consequence. They can be used in the EDA, but consider whether they belong as features.
- **Confounding events:** heavy-vehicle diversion 2018, AP-7 South toll-free 2020, Covid 2020. All fall in the "before" class.
- **Decimal comma** in hour and km_point ("23,33").

## 6. Methodological choices you need to make

| Topic | Options | Questions to consider |
|---|---|---|
| Time window | Full 2010–2023 · symmetric window (e.g. Sep 2018–Aug 2021 vs. Sep 2021–Dec 2023) · exclude 2020 | Does a shorter, symmetric window reduce trend bias? What do you lose? |
| Control group | Former toll roads vs. other interurban roads (difference-in-differences thinking) · parallel roads N-II/N-340 · roads that still have tolls | If the change is just as large on roads that were not affected – what does that say about the cause? |
| Derived features | ex_toll, parallel_road (N-II, N-340, N-240), AP-7 North/South via km_point, motorway based on road prefix (AP-, A-, B-, C-3x), weekend/night interactions | Which features encode the hypothesis directly? |
| Imbalance | Class weights · down-/up-sampling · SMOTE · threshold tuning | Resampling only on training data – why? |
| Validation | Random stratified k-fold vs. time-based split | What does the model learn if training and test data come from the same months? |
| Metrics | ROC-AUC, PR-AUC, balanced accuracy, sensitivity/specificity, F1 | Why is accuracy meaningless at 84/16? |
| Models | Logistic regression (interpretable, odds ratios) · decision tree / random forest / boosting + variable importance | Prediction or explanation – which matters most here? |

# 7. The meeting with Marta – what to bring

She asks for "a rough exploratory data analysis" and questions. The goal is to show that you have understood the problem and the dataset, not to have a finished model.

## Bring (2–4 slides or a short document)

- **Problem statement in your own words**, including why before/after ≠ cause.
- **Dataset overview:** size, period (2010–2023), class balance, variable groups.
- **Figures 1 and 2:** accidents per month with the 1 Sep 2021 line; former toll roads vs. the rest per year.
- **Data quality findings:** road_type inconsistency, 999 codes, NA trends, the "Other units" jump.
- **Preliminary plan:** time window, ex_toll variable, control group, imbalance strategy, validation and metric.

## Questions for Marta

- The assignment says 2010–2024, but the dataset (last updated Dec. 2024) only goes to 2023. Should we use what is available, or is there another file?
- Is it acceptable to restrict the time window (e.g. 2017–2023 without 2020) to reduce trend effects?
- Can we use a control-group approach (former toll roads vs. other roads), or do you want a single global classifier on all accidents?
- Should variables that are clearly time trends (e-scooters / "other units", NA shares) be removed, or is the point precisely to show that the model picks them up?
- Can we use external data (traffic volumes/AADT from Servei Català de Trànsit or the Ministry) to normalise accidents by traffic volume?
- Do you expect the classifier to be validated with a time-based split, or is stratified k-fold sufficient?
- The AP-7 South became toll-free on 1 Jan 2020. Should we separate it out, or treat all former toll roads the same?

## Checklist before the meeting

| | |
|---|---|
| ☐ | 01_first_look.R run, figures made yourselves in R/ggplot2 |
| ☐ | ex_toll variable created from *road* and the road × road_type cross-table checked |
| ☐ | Overview of missing values (incl. 999/Unspecified) per variable and per year |
| ☐ | Before/after shares for 5–6 key variables (accident_type, vehicle types, time of day, zone) |
| ☐ | Group agreed on the questions and who presents what |

# Sources

Figures in sections 4–5 are computed from the dataset (CSV downloaded 15 Oct 2026). Media figures in section 1 are secondary sources – verify against Servei Català de Trànsit before using them in the report.


• Assignment text and presentation (Atenea): TermProject_Option1.pdf, TermProject_presentation.pdf

• Dades Obertes Catalunya – Accidents de trànsit amb morts o ferits greus a Catalunya
https://analisi.transparenciacatalunya.cat/Transport/Accidents-de-tr-nsit-amb-morts-o-ferits-greus-a-Ca/rmgc-ncpb

• 3Cat – Tot el que cal saber sobre la fi dels peatges (2021)
https://www.3cat.cat/324/tot-el-que-cal-saber-sobre-la-fi-dels-peatges-a-part-de-les-autopistes-catalanes/noticia/3115833/

• elDiario.es – Aumenta un 20% el tráfico en la AP-7 sin peajes (1 Oct 2021)
https://www.eldiario.es/catalunya/aumenta-20-trafico-autopista-ap-7-peajes_1_8356255.html

• Ara – Accidents on toll-free motorways increase but fall across the whole network (11 Oct 2021)
https://en.ara.cat/society/accidents-toll-free-motorways-increase-but-fall-across-the-whole-network_1_4145230.html

• Catalan News – Crashes with injuries on AP-7 up 28% since toll removal (7 Oct 2024)
https://www.catalannews.com/traffic/item/crashes-with-injuries-on-ap-7-up-28-since-toll-removal

• Ebre Digital – Es compleix mig any del desviament de camions a l'AP-7 (2019)
https://ebredigital.cat/2019/03/02/es-compleix-mig-any-del-desviament-de-camions-a-lap-7